TRIGONOMETRIC FUNCTIONS

DIFFERENTIAL CALCULUS

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TOPIC OUTLINE

Derivative of Trigonometric Functions

- sine
- cosine
- tangent
- cotangent
- secant
- cosecant

DERIVATIVE OF TRIGONOMETRIC FUNCTIONS

RECIPROCAL AND RATIO IDENTITIES

$$\csc\theta = \frac{1}{\sin\theta}$$

$$\sec\theta = \frac{1}{\cos\theta}$$

$$\cot\theta = \frac{1}{\tan\theta}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

PYTHAGOREAN IDENTITIES

$$\cos^2\theta + \sin^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

SINE

$$\frac{d}{dx}(\sin x) = \cos x$$

Differentiate $y = \sin x + 5$.

$$dy/dx = d/dx(\sin x + 5)$$

$$dy/dx = d/dx(\sin x) + \cancel{d/dx(5)} \to 0$$

$$\boxed{dy/dx = \cos x}$$

ans

SINE

Evaluate $\frac{d}{dx}(3\sin x)$.

constant multiple rule

$$\frac{d}{dx}(3\sin x) = 3\frac{d}{dx}(\sin x)$$

$$\boxed{\frac{d}{dx}(3\sin x) = 3\cos x}$$

ans

$$\frac{d}{dx}(\sin x) = \cos x$$

COSINE

$$\frac{d}{dx}(\cos x) = -\sin x$$

Differentiate $y = \cos x + 5$.

$$dy/dx = d/dx(\cos x + 5)$$

$$dy/dx = d/dx(\cos x) + \cancel{d/dx(5)} \to 0$$

$$\boxed{dy/dx = -\sin x}$$

ans

COSINE

$$\frac{d}{dx}(\cos x) = -\sin x$$

if $u = -\cos x$, find $\frac{du}{dx}$.

$u = -\cos x$

$du/dx = d/dx\,(-\cos x)$

constant multiple rule

$du/dx = -\,d/dx\,(\cos x)$

$du/dx = -(-\sin x)$

$$\boxed{du/dx = \sin x}$$

ans

TANGENT

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

Differentiate $y = \frac{\sin x}{\cos x}$. $\begin{matrix} u \\ v \end{matrix}$

$$dy/dx = d/dx\left(\frac{\sin x}{\cos x}\right)$$

$$dy/dx = \frac{\cos x \; d/dx(\sin x) - \sin x \; d/dx(\cos x)}{(\cos x)^2}$$

Quotient rule

$$\frac{vdu - udv}{v^2}$$

$$dy/dx = \frac{\cos x(\cos x) - \sin x(-\sin x)}{(\cos x)^2}$$

Pythagorean identity

$$dy/dx = \frac{(\cos x)^2 + (\sin x)^2}{(\cos x)^2} \longleftarrow \cos^2 x + \sin^2 x = 1$$

TANGENT

Differentiate $y = \frac{\sin x}{\cos x}$. u v

$dy/dx = \cancel{\frac{1}{(\cos x)^2}} \rightarrow \frac{1}{\cos x} = \sec x$

$\boxed{dy/dx = \sec^2 x}$

ans

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

COTANGENT

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

Differentiate $y = \frac{\cos x}{\sin x}$. $\begin{matrix} u \\ v \end{matrix}$

$$dy/dx = d/dx\left(\frac{\cos x}{\sin x}\right)$$

$$dy/dx = \frac{\overset{v}{\sin x}\,\overset{du}{d/dx(\cos x)} - \overset{u}{\cos x}\,\overset{dv}{d/dx(\sin x)}}{(\sin x)^2 \quad v^2}$$

$$dy/dx = \frac{\sin x(-\sin x) - \cos x(\cos x)}{(\sin x)^2}$$

$$dy/dx = \frac{-(\sin x)^2 - (\cos x)^2}{(\sin x)^2}$$

COTANGENT

Differentiate $y = \frac{\cos x}{\sin x}$. $\begin{matrix} u \\ v \end{matrix}$

$$\frac{dy}{dx} = \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x}$$ ← Pythagorean identity

$$\frac{dy}{dx} = -\frac{1}{\sin^2 x}$$ → $\frac{1}{\sin x} = \csc x$

$$\boxed{\frac{dy}{dx} = -\csc^2 x}$$ ans

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

SECANT

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

Differentiate $y = \frac{1}{\cos x}$. $\begin{matrix} u \\ v \end{matrix}$

$$dy/dx = d/dx\left(\frac{1}{\cos x}\right)$$

$$dy/dx = \frac{\overset{v}{\cos x}\ \overset{du \to 0}{\cancel{d/dx(1)}} - \overset{u}{1}\ \overset{dv}{d/dx(\cos x)}}{(\cos x)^2}$$

$$dy/dx = \frac{\cancel{\cos x(0)} - (-\sin x)}{(\cos x)^2}$$

$$dy/dx = \frac{\sin x}{\cos x \cdot \cos x} \rightarrow \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x}$$

SECANT

Differentiate $y = \frac{1}{\cos x}$. $\begin{matrix} u \\ v \end{matrix}$

$$\frac{dy}{dx} = \overset{\sec x}{\frac{1}{\cos x}} \cdot \overset{\tan x}{\frac{\sin x}{\cos x}}$$

$$\boxed{\frac{dy}{dx} = \sec x \tan x}$$

ans

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

COSECANT

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

Differentiate $y = \frac{1}{\sin x}$. $\begin{matrix} u \\ v \end{matrix}$

$$dy/dx = \frac{d}{dx}\left(\frac{1}{\sin x}\right)$$

$$dy/dx = \frac{\overset{v}{\sin x}\, \overset{du \to 0}{\cancel{\frac{d}{dx}(1)}} - \overset{u}{1}\, \overset{dv}{\frac{d}{dx}(\sin x)}}{(\sin x)^2 \quad v^2}$$

$$dy/dx = \frac{\cancel{\sin x}(\overset{\to 0}{0}) - (\cos x)}{(\sin x)^2}$$

$$dy/dx = \frac{-\cos x}{\sin x \cdot \sin x} \rightarrow \frac{-1}{\sin x} \cdot \frac{\cos x}{\sin x}$$

COSECANT

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

Differentiate $y = \frac{1}{\sin x}$. $\frac{u}{v}$

$$\frac{dy}{dx} = -\overset{\csc x}{\cancel{\frac{1}{\sin x}}} \cdot \overset{\cot x}{\cancel{\frac{\cos x}{\sin x}}}$$

$$\boxed{\frac{dy}{dx} = -\csc x \cot x}$$

ans

EXERCISE

Differentiate $y = x^2 \sin x$.

Solution

$$\frac{dy}{dx} = \frac{d}{dx}(\overset{u}{x^2}\,\overset{v}{\sin x})$$

$$\frac{d}{dx} = \overset{v}{\sin x}\,\overset{du}{\frac{d}{dx}}(x^2) + \overset{u}{x^2}\,\overset{dv}{\frac{d}{dx}}(\sin x)$$

$$\frac{d}{dx} = \sin x\,(2x) + x^2(\cos x)$$

$$\boxed{\frac{d}{dx} = 2x\sin x + x^2\cos x}$$

ans

EXERCISE

Differentiate $y = \sqrt{x}\sin x$.

$$\frac{dy}{dx} = \frac{d}{dx}\left(\overset{u}{x^{\frac{1}{2}}}\,\overset{v}{\sin x}\right)$$

$$\frac{dy}{dx} = \overset{v}{\sin x}\,\overset{du}{\frac{d}{dx}\left(x^{\frac{1}{2}}\right)} + \overset{u}{x^{\frac{1}{2}}}\,\overset{dv}{\frac{d}{dx}(\sin x)}$$

$$\frac{dy}{dx} = \sin x\left(\frac{1}{2}x^{-\frac{1}{2}}\right) + x^{\frac{1}{2}}(\cos x)$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}\sin x + \sqrt{x}\cos x$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}} \cdot \frac{\sqrt{x}}{\sqrt{x}}\sin x + \sqrt{x}\cos x$$

Solution

$$\frac{dy}{dx} = \frac{\sqrt{x}}{2x}\sin x + \sqrt{x}\cos x$$

$$\boxed{\frac{dy}{dx} = \sqrt{x}\left(\frac{\sin x}{2x} + \cos x\right)}$$

ans

EXERCISE

Differentiate $y = 3x^2 - 2\cos x$.

Solution

$$\frac{dy}{dx} = \frac{d}{dx}(3x^2) - \frac{d}{dx}(2\cos x)$$

$$\frac{dy}{dx} = 3\frac{d}{dx}(x^2) - 2\frac{d}{dx}(\cos x)$$

$$\frac{dy}{dx} = 6x - 2(-\sin x)$$

$$\boxed{\frac{dy}{dx} = 6x + 2\sin x}$$

ans

EXERCISE

If $u(x) = \frac{\cos x}{1-\sin x}$, find $\frac{du(x)}{dx}$.

Solution

$$\frac{d}{dx}\, u(x) = \frac{d}{dx}\left(\frac{\overset{u}{\cos x}}{\underset{v}{1-\sin x}}\right)$$

$$\frac{d}{dx}\, u(x) = \frac{\overset{v}{1-\sin x}\;\overset{du}{\frac{d}{dx}(\cos x)} - \overset{u}{\cos x}\;\overset{dv}{\frac{d}{dx}(\cancel{1}^{\to 0}-\sin x)}}{(1-\sin x)^2 \quad v^2}$$

$$\frac{d}{dx}\, u(x) = \frac{(1-\sin x)(-\sin x) - \cos x\,(-\cos x)}{(1-\sin x)^2}$$

EXERCISE

If $u(x) = \frac{\cos x}{1-\sin x}$, find $\frac{du(x)}{dx}$.

Solution

$\sin^2 x + \cos^2 x = 1$

$$\frac{d}{dx}u(x) = \frac{(-\sin x + \sin^2 x) + (\cos^2 x)}{(1-\sin x)^2}$$

$$\frac{d}{dx}u(x) = \frac{1-\sin x}{(1-\sin x)^2} \leftarrow \frac{\cancel{1-\sin x}\ 1}{\cancel{(1-\sin x)}(1-\sin x)}$$

$$\boxed{\frac{d}{dx}u(x) = \frac{1}{1-\sin x}}$$

ans

EXERCISE

Evaluate $\frac{d}{d\theta}\left(\frac{\sec\theta}{1+\sec\theta}\right)$.

Solution

$$y = \frac{\overset{u}{\sec\theta}}{\underset{v}{1+\sec\theta}}$$

$$\frac{dy}{d\theta} = \frac{\overset{v}{(1+\sec\theta)}\ \overset{du}{\frac{d}{d\theta}(\sec\theta)} - \overset{u}{\sec\theta}\ \overset{dv}{\frac{d}{d\theta}(\cancelto{0}{1}+\sec\theta)}}{\underset{v^2}{(1+\sec\theta)^2}}$$

$$\frac{dy}{d\theta} = \frac{(1+\sec\theta)(\sec\theta\tan\theta) - \sec\theta(\sec\theta\tan\theta)}{(1+\sec\theta)^2}$$

$$\frac{dy}{d\theta} = \frac{(\sec\theta\tan\theta + \cancel{\sec^2\theta\tan\theta}) \cancel{-} (\cancel{\sec^2\theta\tan\theta})}{(1+\sec\theta)^2}$$

$$\boxed{\frac{dy}{d\theta} = \frac{\sec\theta\tan\theta}{(1+\sec\theta)^2}}$$

ans

EXERCISE

Differentiate $g(t) = 4\sec t + \tan t$.

Solution

$$g(t)' = \frac{d}{dt}(4\sec t + \tan t)$$

$$g(t)' = 4\frac{d}{dt}(\sec t) + \frac{d}{dt}(\tan t)$$

$$g(t)' = 4(\sec t \tan t) + (\sec^2 t)$$

$$\boxed{g(t)' = 4\sec t \tan t + \sec^2 t}$$

ans

EXERCISE

Differentiate $f(\theta) = \dfrac{\overset{u}{\sec\theta}}{\underset{v}{1 + \tan\theta}}$.

Solution

$$f(\theta)' = \frac{\overset{v}{(1+\tan\theta)}\,\overset{du}{\frac{d}{d\theta}(\sec\theta)} - \overset{u}{\sec\theta}\,\overset{dv}{\frac{d}{d\theta}(\cancel{1}\overset{\to 0}{+}\tan\theta)}}{(1+\tan\theta)^2} \quad v^2$$

$$f(\theta)' = \frac{(1+\tan\theta)(\sec\theta\tan\theta) - \sec\theta(\sec^2\theta)}{(1+\tan\theta)^2}$$

$$f(\theta)' = \frac{(\sec\theta\tan\theta + \sec\theta\tan^2\theta) - (\sec^2\theta)}{(1+\tan\theta)^2}$$

EXERCISE

Differentiate $f(\theta) = \dfrac{\sec\theta}{1+\tan\theta}$. u v

Solution

$$f(\theta)' = \frac{\sec\theta(\tan\theta + \cancel{\tan^2\theta - \sec^2\theta})}{(1+\tan\theta)^2}$$ -1

Pythagorean Identity

$1 + \tan^2\theta = \sec^2\theta$

then, $\tan^2\theta - \sec^2\theta = -1$

$$\boxed{f(\theta)' = \frac{\sec\theta(\tan\theta - 1)}{(1+\tan\theta)^2}}$$

ans

EXERCISE

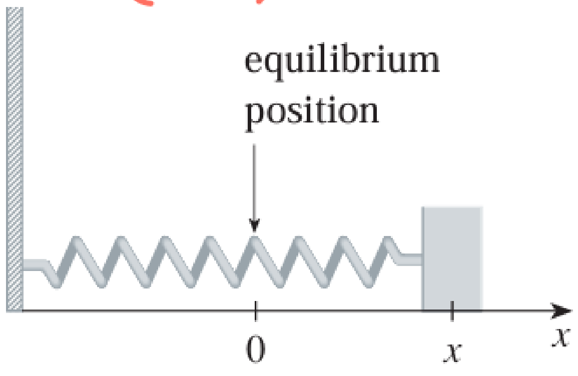
A mass on a spring vibrates horizontally on a smooth level surface (see figure). Its equation of motion is $\boldsymbol{x}(\boldsymbol{t}) = \mathbf{8}\sin \boldsymbol{t}$, where t is in seconds and x in centimeters.

$\frac{d}{dt}x(t)$

a. Find the velocity at time t

b. Find the position of the mass at time $t = \frac{2\pi}{3}$

$x(2\pi/3)$



Solution

Velocity

$x(t) = 8\sin t$

$\frac{d}{dt}x(t) = \frac{d}{dt}(8\sin t)$

$$\boxed{\frac{d}{dt}x(t) = 8\cos t}$$

ans

EXERCISE

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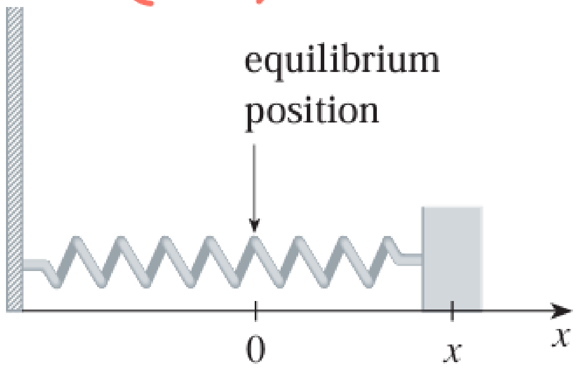

Solution

Position at $t = \frac{2\pi}{3}$ sec

$$\frac{d}{dt}x(t) = 8\cos t$$

$$\frac{d}{dt}x(2\pi/3) = 8\cos(2\pi/3)$$

$$\boxed{\frac{d}{dt}x(2\pi/3) = -4\text{ cm}}$$

ans

note: angle in radian

LABORATORY